Engineering Mathematics - I · Unit_05 · MATLAB

DRAFT — UE26MA141A · Unit_05 · MATLAB · Introduction to MATLAB Software

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Introduction to MATLAB Software

1 Introduction to MATLAB Software

Study focus: Introduction to MATLAB Software. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Sibashankari from the Department of Science and Humanities.

Key details

- For the semester, in the lab session, we will cover basics of MATLAB, programming in MATLAB, plotting 2D and 3D, and then linear algebra, partial differentiation, and solving differential equation.
- And the reference books are "Getting Started with MATLAB" by Rudra Pratap, and the "MATLAB for Engineers" by Holly Moore, and then "MATLAB Programming for Engineers" by Stephen J.
- The whole syllabus will be covered in 12 sessions, and this is the first session in MATLAB.
- Hope everyone would have downloaded the MATLAB software.
- In case if you have not downloaded, then you can use the installation guidelines link that is attached along with this video.
- As that MATLAB is a paid software, and so if you have institution mail.
- ID, then you can get the access to licensed version.
- Otherwise, you need to download the trial version.
- MATLAB is a software package for numerical computation, visualization, and programming.
- It provides an interactive environment with hundreds of built-in function for technical computation, graphics, and animation.
- The name MATLAB stands for Matrix Laboratory because the basic building block of MATLAB is the matrix.
- Some basic features and commands of MATLAB.
- It consists of MATLAB desktop, a figure window, and an editor window.

- The figure window opens automatically when graphic commands are executed and contains graph created by these command.
- The editor window is used for writing and editing programs.
- This is the place where you can write, edit, create, or save your own program in files called M files.
- We can see the MATLAB desktop, which contains three sub-windows, command window, current folder, and the workspace.
- This is the default layout.
- If you want to change the layout, click on the layout option here and change as per your preference.
- Other two windows are not displayed.
- This is the main window where we write the commands as well as see the outputs.
- In other words, here is the interaction with the software.
- So to type a command, the cursor must be placed next to the command prompt.
- To start with, we can use this command window as a calculator where you can type in commands line by line.

Example or application from the lecture

- To begin with, let us look at the general structure of MATLAB environment.
- Suppose if I select the three column, then the current folder, command window, and the workspace appear in column-wise.
- Suppose if I select it as command window only, then you could see only the command window.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

Once a command is typed, to execute the command, Enter key is pressed.

Key details

- Whenever a calculation is performed, MATLAB will assign the result to the built-in variable ans, unless specified.
- That is the answer variable.
- And ans is the default output variable.
- The easy way to do is, is to use an assignment operator.
- Use the assignment operator, which is equal sign, and then any value or mathematical expression we can give.
- And you should remember that, or make a note of it, the variable must be always on the left, and the expression or the value should be on the right.
- Otherwise, an error will occur if they are reversed.
- And one more thing is that if I type b plus four equal to 66, see, this is an incorrect assignment statement.
- It takes small a and capital A as two different variable.

- That is, the variable should not start with a number.
- It should always start with the alphabet.
- I cannot assign it this way.
- Instead, if I assign a4 equal to 20, it will accept.
- It should start with alphabet.
- And we should avoid using the names of built-in function for variable.
- As my variable, sin equal to 66, then sin variable is created and the value 66 is stored in that sin variable.

Example or application from the lecture

- And after alphabet, a number can be given, any symbol can be given, even underscore can be given, but a space should not be there.
- Suppose if I use sin.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

And how do we know that it is a built-in function?

Key details

- This will give you what are the variables that are there in the workspace.
- Or if you want to know about the size of the data type, then you can just type whos.
- Then it will give you the variable along with the size and the data type.
- And coming to the next one, several commands can be typed in the same line.
- That is, I want to find out the sum of it, then I want to find out the difference, I want to find out a/b, then I want to find out a star b. a star b indicates the multiplication.
- So first this is a+b, and this is a-b, then this is a/b, and this is a*b.
- I'll give it as 33, then, and I don't want to display every time when I give a equal to 33, and when I press the enter key, that value is displayed.
- If I don't want to see that it is being displayed on the screen, what we can do is that for assigning b value, so b equal to, say, if I write 44, I can put one semicolon.
- What is this semicolon will do is that it will suppress the output.
- But in the workspace, if you see, b takes the value 44.
- It is stored there.
- So either what we can do is that you can copy-paste those operation, or you can keep the cursor next to the command prompt and click the up arrow key in your keyboard.
- Then you can see that command.
- You can recall that command.
- So first is addition, subtraction, then I will do multiplication, then I will do the division.
- This is a+b, a-b, a*b, and a/b.

Example or application from the lecture

- Suppose when you are assigning a value to your variable and if you want to check whether you have already used this variable name, or if you want to have the list of variables used, just type who.
- Suppose if you want to modify or make a correction, you can do it.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

We will do some arithmetic operation in MATLAB.

Key details

- Percentage symbol, arithmetic operations.
- Usually, there's no need for commands in the command window, but this is very useful when we are doing programs.
- We will take first problem, and we will execute this.
- Calculate the following using MATLAB command: one by two plus three square, plus four by five into six by seven.
- So actually, we have to do term by term, first is one by two plus three square, and then four by five into six by seven.
- And for multiplication, I should use asterisk symbol six by seven.
- Usually, MATLAB by default displays number with four decimal points.
- If you do so, now if you calculate any problem, say two star sin of 1.4.
- Take on the next problem.
- Find the value of Y equal to E power minus A sine X plus 10 root Y, if A equal to 5, X is equal to 2, and Y is equal to 8.
- So first step, we have to declare these variables.
- A is 5.
- A equal to 5.
- X equal to 2.
- And then Y is equal to 8.
- I need to find out Y is equal to E power minus A sine X plus 10 root Y.

Example or application from the lecture

- The first preference is given to the parenthesis, next is the exponential, then comes the multiplication and division, and finally addition and subtraction.
- Suppose if you see multiplication and division or addition and subtraction on the same line or on the simultaneously if it is present, then it is solved from left to right.
- Suppose if you want to be more accurate, then what you can do is that you can type format long.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

plus 10 into square root of Y.

Key details

- Compute sine pi by 4 and 4 into E power 10 sine 30 degree cos inverse pi by 4 and log 10 of exponential E power 2 plus cos pi.
- First, we will do sine pi by 4.
- Don't use the uppercase.
- The next is four into exponential.
- E power 10.
- For exponential, you have to type it as the inbuilt function as exp of 10.
- And if you observe this, the answer is 8.8106 into N power 4.
- So that what it this means.
- Now the next is, we need to find out the value of sine 30 degree.
- You cannot find out for 30 degree.
- You can only find out for the radian.
- that to convert the degree into radian, you have to multiply by pi by 180.
- You need not convert it to radian mode.
- Now coming to the next problem, we need to find out cos inverse of pi by 4.
- This also it is there in the PPT.
- And pi by 4, as I have told you before, pI is the built-in variable by 4.
- And the last problem, log 10 of E power 2 plus cos pi.
- And always that built-in function has to be within the open brackets.

Example or application from the lecture

- Suppose if you want to do only in the degree mode, then you have to use another built-in function, which is S-I-N-D.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 25:00–29:59

into of E, exponential is exp, 2 plus, cos is also a built-in function, I need to open the bracket, pI.

Key details

- Hope everyone would have got some idea how to work with MATLAB.
- Practice those problems and make yourself familiar with MATLAB interface.
- So we will meet you in the next session.

Example or application from the lecture

- Some more problems are given in the assignment section.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Introduction to MATLAB Software transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to MATLAB Software transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to MATLAB Software transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Introduction to MATLAB Software transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 4 rendered pages in 001_Slides__7ac00f797de85ee4.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 002_Slides__ef919841425356a4.slides.md have no readable `.slides.md` text; their original page images are included.
3. 1 rendered pages in 003_Slides__3d7af4a88d5b4490.slides.md have no readable `.slides.md` text; their original page images are included.
4. 1 rendered pages in 004_Slides__4150cd4c33be9a2c.slides.md have no readable `.slides.md` text; their original page images are included.
5. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
6. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
7. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
8. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
9. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
10. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
11. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00002	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare...
SRC00003	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare...
SRC00005	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare...

C Revision and retrieval practice